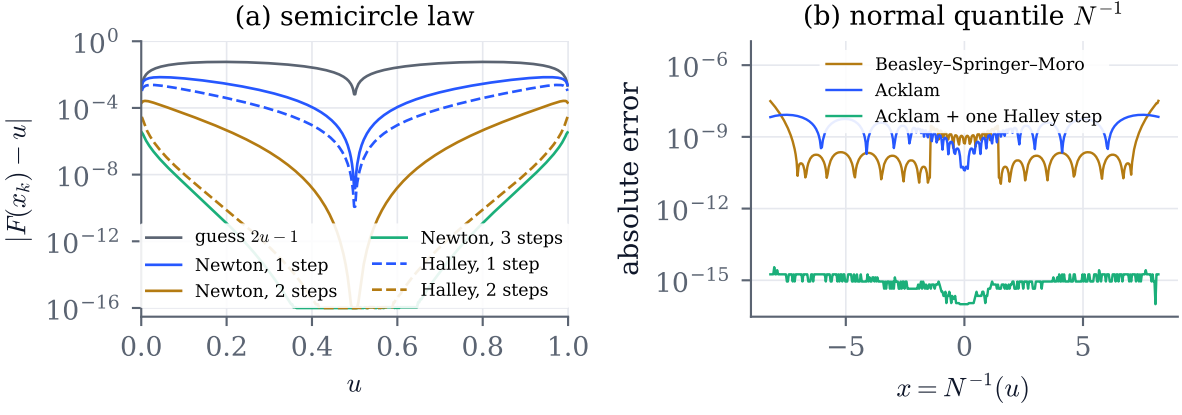
(a) semicircle law
10^{0}
10^{-4}
10^{-8}
10^{-12}
10^{-16}
$|F(x_k) - u|$
guess $2u - 1$
Newton, 1 step
Newton, 2 steps
Newton, 3 steps
Halley, 1 step
Halley, 2 steps
0.0
0.2
0.4
0.6
0.8
1.0
u
(b) normal quantile N^{-1}
10^{-6}
10^{-9}
10^{-12}
10^{-15}
absolute error
Beasley–Springer–Moro
Acklam
Acklam + one Halley step
−5
0
5
$x = N^{-1}(u)$